Projects for the lecture Intelligent User Interfaces

UNIVERSITÄT BAYREUTH

Title. Useful elements:

- A short title, e.g. a project name or "product" name
- A telling title below, which shortly summarises the key elements of approach and goal

PosterMaker

ng interactive layout generation to enable students to create better posters

How might we make layout generation interactively usable to support students in creating better posters?
Our application "PosterMaker" uses Deep Learning to extract commonly used elements from a large dataset of example posters. We apply this model in a user interface that suggests possible further eleme design. With our conc many poster variation and creative poster de

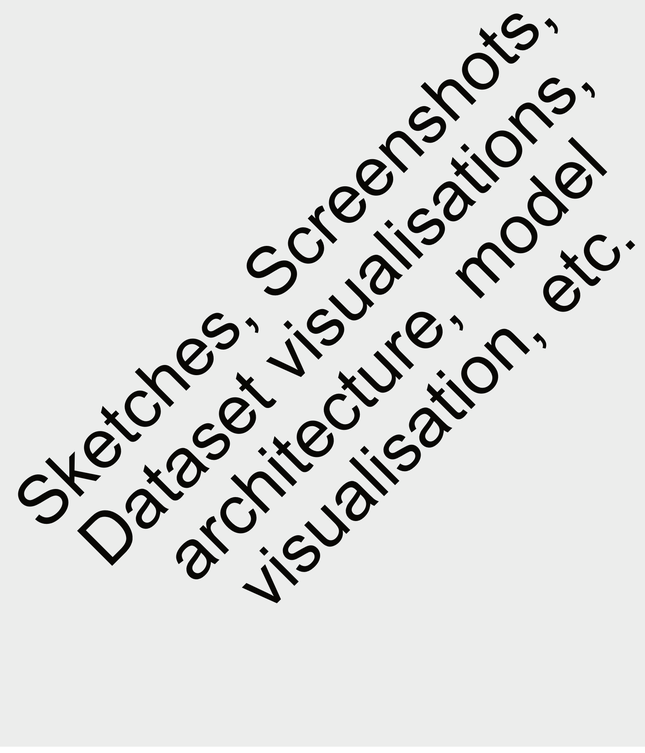


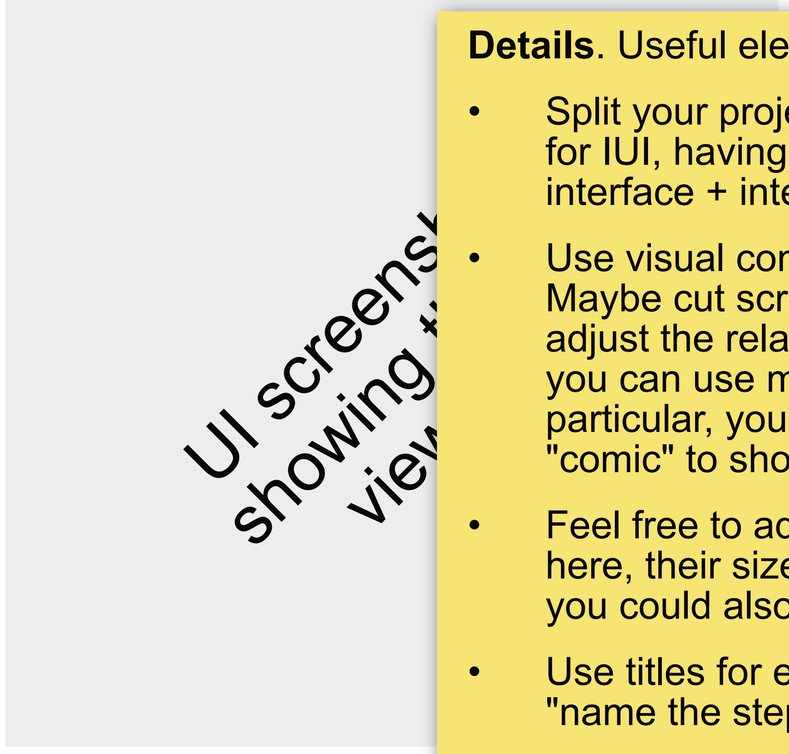
Abstract (short summary) of the project. Useful elements:

- What's the problem / goal? (e.g. as a "how might we" question)?
- What you have done / built (e.g. methods/approach, created system; be specific but don't put in too many details)?
- What is the result / impact (e.g. how your system is useful to people)?

Section header. Feel free to adjust, sets the context to the text and images below.

Overview of our project:





Details. Useful elements / hints:

- Split your project and approach into a few parts (e.g. 3); for IUI, having sth. about technical foundation + user interface + interaction is a good default split
- Use visual content (screenshots, visualisations, etc.). Maybe cut screenshots to the relevant parts. Feel free to adjust the relative size and number of elements (e.g. you can use more images than the 3 boxes here). In particular, you might need a series of screenshots like a "comic" to show how the interaction works.
- Feel free to adjust the layout, e.g. the number of parts here, their sizes, etc. If you have very broad images, you could also put the parts in a vertical layout.
- Use titles for each part - the example has this pattern: "name the step/part": "What you have done"
- Keep the text short; really think about what the key info is for the readers to understand your project overall. It should be possible for people to read the steps in 1-2 minutes max.

Technical foundation: Representation learning on a large poster dataset

We applied a deep auto encoder to an existing dataset of poster layouts, to learn a latent representation of the design space of scientific posters.

The image above visualises a 2D embedding of the resulting learned space. Some patterns are directly visible: For example, we identified clusters of major design choices (e.g. text-heavy vs visual posters).

UI concept: Auto-visual poster editi

We employed our model in a user interface concept which builds on well-known interfaces for visual editing (e.g. Photoshop): The central element is a canvas with a toolbar.

Users can manually place text and images on the canvas. At the same time, the model analyses the current design to suggest possible elements to add. These suggestions are shown in light blue. Users can ignore them, or accept them with one click.

In the first example above, the user has placed a header and a main text element. As shown, the system now suggests two elements: a footer and an image element.

In the second image, the user has clicked on the image element, to accept this suggestion. The UI has changed: The system suggests to add text next to this new image element.

Now, the user decides to manually place another element, which again updates the suggestions.

Footer. Team members (name, maybe study program); other info (e.g. link to a project website, online prototype, etc.)

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